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Rapid bidirectional prediction between physical field and key control parameters in tunnel fires

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A R T I C L E I N F O

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A B S T R A C T

Tunnel fire poses a serious threat to social public safety, and the losses they cause are often incalculable. The prediction of tunnel fires contributes to decision-making in rescue and firefighting, and helpfully reduces fire losses as much as possible. The financially expensive experiments and the time-consuming simulation slow down the pace of development in tunnel fire prediction. Moreover, numerical and experimental study is often unidirectional, with the characteristic of predicting less dimensional data through higher dimensional data. This work proposes a deep learning model (DLM) to instantly achieve bidirectional prediction between the full field information of tunnel fires and a small amount of key physical quantities. Under the designed data processing method, the DLM is trained by a big tunnel fire numerical database with various ventilation, thermal, and geometric conditions. The results show that the DLM can learn the physical fields data and the physical quantities data well with the increasing training epoch. In addition, the DLM performs the promising bidirectional prediction. From the symmetry comparison, the result shows the full physical fields are well predicted by the decoder part of DLM via four key physical quantities. The prediction of the key physical quantities is overall satisfactory, but the prediction accuracy of the tunnel inclination angle is relatively poor compared with the other quantities. The prediction accuracy of key physical parameters through the temperature field is better than through smoke visibility. The important parameters in practice, namely smoke layer distribution and smoke back-layering length are also predicted, and the R^2 of 0.95 and 0.92 are respectively obtained. The bidirectional prediction system proposed in this work demonstrates the promising application for intuitive and rapid prediction of various information in tunnel fires, as well as for summaries of physical laws in tunnel fires.

1. Introduction

It is almost impossible as of now to have high-speed public transportation, exploration of massive energy, and construction of lifelines without the establishment of tunnels. Consequently, the safety and security of the tunnel are increasingly crucial. Vehicle accidents, fuel leaks, and other issues are likely to trigger the fire in long-narrow spaces and have deadly consequences. Fire as one of the destructive disasters in tunnels therefore cannot be underestimated due to its huge threat to lives and properties. The hazards of tunnel fires are mainly reflected in the toxicity of smoke and the destructive power of heat. In fact, the movement of smoke is driven by thermal buoyancy, so the smoke and heat normally harm people and properties meanwhile in tunnels. To reduce the losses caused by fires and help rescue decision-making, the physical laws of tunnel fires (Beard and Carvel, 2012; Carvel, 2019; Casey, 2020; Ingason et al., 2015; Li and Liu, 2020) were widely studied.

The prediction of tunnel fires is thus an important research line with the study methodologies including numerical simulation (Meng et al., 2014), theoretical analysis (Hong and Fu, 2022), as well as experiments combined with dimensional analysis (Du et al., 2022). By means of the experimental investigation, some key parameters in tunnel fires such as the critical velocity (Liu et al., 2020), back-layering length (Hu et al., 2008), and the maximum temperature beneath the tunnel ceiling (Li et al., 2011) have been thoroughly studied. In addition, by combining with the dimensional analysis, many excellent correlations were proposed by fitting the coefficients in the dimensionless correlations (Gao et al., 2022; Huang et al., 2023). Early, due to the lack of theoretical physical analysis in some of the correlations, those correlations were phenomenological (Wu and Bakar, 2000). Hence, more and more theoretical analyses followed since then, which enhanced the physical explanations and understanding of the calculation correlations (Wang

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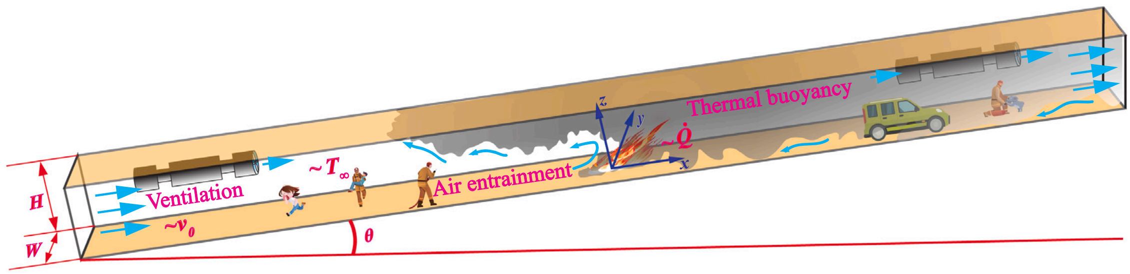

Fig. 1. Schematic diagram of a tunnel fire scenario.

et al., 2017; Kunsch, 2002). In short, the theoretical and experimental results provide convenient tools for predicting some key parameters in tunnel fires. However, the conduction of experiments is financially expensive. On the other hand, these key parameters predicted by the correlations cannot offer the details of the entire physical field in the tunnel fires, which hinders the analysis of accidents and the prevention of future disasters. The development of Computational Fluid Dynamics (CFD) contributes to the numerical simulation technology for fire-related studies, which could computationally give the full field information step-by-step (Hong et al., 2022b). With the rapid increase in computing power, numerical simulation technology has been able to substitute many experimental works at an affordable cost. Especially for those turbulent models in CFD that support coarser grid cell size such as the Large Eddy Simulation (LES) and Reynolds-Averaged Navier–Stokes (RANS) model, though losing some acceptable accuracy in engineering. Through CFD, tunnel fires can be intuitively predicted, such as plug-holing phenomena in vertical ventilation systems and air-entrainment phenomena (Wang et al., 2022; Chen et al., 2023). Moreover, the geometric dimensions, structure of tunnels, and physical conditions can be easily changed and paralleled computed by CFD (Hong et al., 2022a), providing a rapid and theoretical basis for the fire protection design of tunnels. Although CFD saves time and costs, its calculation time is still relatively long if we hope to predict an actual fire instantly.

Nowadays, machine learning has made significant progress due to the development of deep neural network models, and has recently been used to aid in the prediction of tunnel fires. Zhang et al. (2021) and Hong et al. (2022a) used machine learning predicted some key parameters in the tunnel first via the experimental data and numerical data, respectively. Hu et al. (2023) used the machine learning model for the prediction of maximum ceiling temperature in tunnel fires, and Sun et al. (2022) predicted the whole ceiling temperature distributions. He et al. (2023) predicted the temperature evolution of tunnel fires by using the machine learning method as well. These predictions can give a key physical scalar or vector prediction of tunnel fires instantly, and their machine learning models were trained by the time-averaged data which is highly significant in the statistical scope. Nevertheless, similar to the correlations, their models cannot give full-field detailed information. Li et al. (2021) proposed a deep learning model (DLM) for forecasting full-field smoke spread in tunnel fires. Zhang et al. (2022) proposed a dual-agent DLM for predicting the full-field information in real-time. Wu et al. (2020) applied the machine learning method of LSTM for predicting the fire locations and other key parameters through the data collected by the sensors installed in the tunnel. These models can predict the full-field information, but the issue is that they focused on the prediction in real time. Actually, the input of the models detected in the tunnel is not real-time in real-world measurements, so their practical significance may not be promising. In fact, considering the robustness in practice, using time-averaged results for predicting full-field information is more statistically significant, similar to the correlation equations mentioned above. Moreover, there is still an issue that the above studies are all unidirectional predictions, either predicting key data using full-field information or predicting full-field information using monitored key data. Therefore, this work aims to attempt to address the issues and fill the gaps mentioned above.

In summary, this paper attempts to establish a system for the instant bidirectional prediction between the full physical fields and some key physical quantities in tunnel fires. To achieve this goal, a deep learning model based on the idea of encoder–decoder has been developed. To practically improve the statistical significance of the proposed system, the prediction result and its input are time-averaged and are based on our large high-resolution numerical database of tunnel fires. A bidirectional prediction of the tunnel fire fields with various tunnel cross-section sizes, ventilation velocities, heat release rates (HRR), tunnel inclinations, and ambient temperatures was demonstrated.

2. Fire scenario

The tunnel fire scenario in this paper is as shown in Fig. 1, where the dimension of the tunnel is H m $\times$ W m $\times$ 60 m. The establishment of the Cartesian coordinate system and its origin is shown at the fire source location in the figure. The tunnel is assumed to have no bifurcation and the walls have no curvature along the x, y, and z axes. The tunnel is equipped with a longitudinal ventilation system while without a transverse ventilation system. Four key physical quantities, inclination angle (θ), HRR ($\dot{Q}$), ventilation velocity (v_0), and ambient temperature (T_∞), are the unknown variables that control the development and the scale of the tunnel fire. The walls of the tunnel are assumed to be composed of concrete. The fire source is assumed to occur on tunnel floor center, where the combustion area of the fire source is 1×1 m^2. Propane is used as the fuel for combustion.

3. Methodology

3.1. Dataset

The dataset used in this work contains 1000 sets of simulation results of tunnel fires. Each case was simulated under six different geometric and physical conditions including v_0, T_∞, $\dot{Q}$, θ, H and W. The six conditions were randomly (uniform distributions are satisfied) sampled in the 1000 sets of simulation. The sampling ranges of the six conditions are: $v_0 \in [0.2, 5]$ m/s, $T_\infty \in [10, 30]$ °C, $\dot{Q} \in [0.35, 5]$ MW, $\theta \in [-15, 15]$ °, $H \in \{2, 2.5, 3, 3.5, 4\}$ m, and $W \in \{3, 3.5, 4, 4.5, 5\}$ m, respectively. It is noted that the ranges are merely references referred to literature of Hong et al. (2022a), and can be expanded or reduced in the future.

The CFD simulations were performed with Fire Dynamics Simulator (FDS), version 6.7.1 (McGrattan et al., 2013). Unless specified otherwise, the default settings have been applied. The specific heat, thermal conductivity, and heat transfer thickness of the material of concrete are set to 1.04 kJ/(kg K) K, 1.8 W/(m K), and 2 m, respectively. The steady-state and uniformly distributed velocity boundary condition is employed to the entire cross-section at the entrance of the tunnel, with a magnitude of v_0. The pressure boundary condition at the tunnel exit is the "OPEN" (constant dynamic pressure) (McGrattan et al., 2013). To achieve a quasi-steady state with statistical significance, the numerical

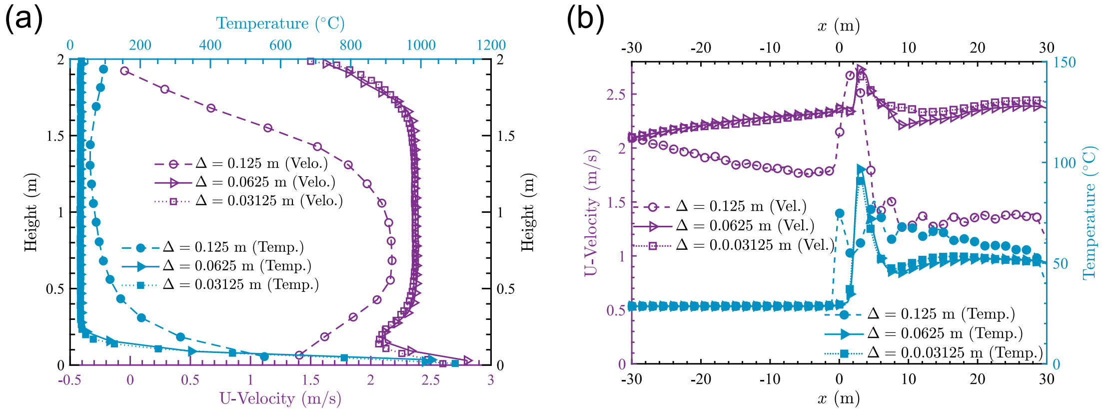

Fig. 2. Temperature and velocity profiles at the lines (a) '$x = 0$ m, $y = 0$ m, 0 m $\leq z \leq 2$ m' and (b) '$-30 \leq x \leq 30$ m, $y = 0$ m, $z = 0$ m'.

simulation time is set to end at 300 s. For the fuel of propane, the yield of CO (y_{CO}), the yield of soot (y_{Soot}), and volume fraction of hydrogen in the soot (X_H) in this work are set to 0.02, 0.01, and 0.1, respectively (Mulholland et al., 1991). As the experimental result reported by Hurley et al. (2015), the radiation fraction for the fuel propane is fixed at 30%.

In order to ensure the precision of the LES result, this work follows the D^* criterion (McGrattan et al., 2013) for determining the grid resolution. D^* criterion considered the mesh has a high resolution when the value of D^*/Δ is close to or higher than 10 (McGrattan, 2007), where $D^* = \left(\frac{\dot{Q}}{\rho_\infty c_p T_\infty \sqrt{g}}\right)^{\frac{2}{5}}$, $\Delta = \sqrt[3]{\delta x \delta y \delta z}$. Here ρ_∞ is the density of the ambient air, g is the gravitational acceleration, δx is the length of grid cells in the x direction, δy is the length of grid cells in the y direction, and δz is the length of grid cells in the z direction. This work used the cubic grid cells, so $\Delta = \delta x = \delta y = \delta z$. In this work, D^* takes the minimum value ($D^*_{\min}$) when the HRR is set to its minimum value ($\dot{Q} = 350$ kW) and ambient temperature is set to the maximum value (30 °C). Therefore, the grid cell size Δ in all 1000 cases can satisfy $D^*/\Delta > 10$ if $\Delta < D^*_{\min}/10 = 0.621$ m. As a result, the grid sensitivity analysis was performed by testing three different values of Δ which are equal to 0.125 m, 0.0625 m, and 0.03125 m, respectively. Fig. 2(a) shows the temperature profile and the x-axis component of the velocity profile along the line segment of '$x = 0$ m, $y = 0$ m, 0 m $\leq z \leq 2$ m' which is the center line above the fire source. The profiles are the time-averaged result between 220 s to 300 s. It is seen that the profiles agree well for the grid cell size of 0.0625 m and 0.03125 m, whereas significant deviations are shown between the profiles calculated by the grid cell size of 0.125 m and the profiles calculated by the other two. Fig. 2(b) illustrates the temperature and velocity distributions at the line segment of '-30 $\leq x \leq$ 30 m, $y = 0$ m, $z = 0$ m' which is the longitudinal center line along the whole tunnel. A similar result is obtained, even though some small differences can also be seen at x = 10 m. The slight deviation is generally acceptable for the large eddy simulations. Therefore, the grid cell size of 0.0625 m is accordingly chosen since it can reduce calculation time while ensuring precision.

Besides, Fig. 3 compares the typical parameter of critical longitudinal ventilation velocity in tunnel fires, including the experimental data from Li et al. (2010), correlation recommended by NFPA 502 (NFPA, 2020), as well as those simulation data with the smoke back-layering equal to zero. It is seen that the simulation data agrees well with the experimental data and the recommendation correlation. The results support again the verification of the simulation dataset.

Tunnel fires are the physical and chemical processes involved with chemical reaction flow. The strong turbulence in the flow field leads to many unstable transient phenomena such as flame oscillation. Therefore, our dataset utilizes the time-averaged results under quasi-steady

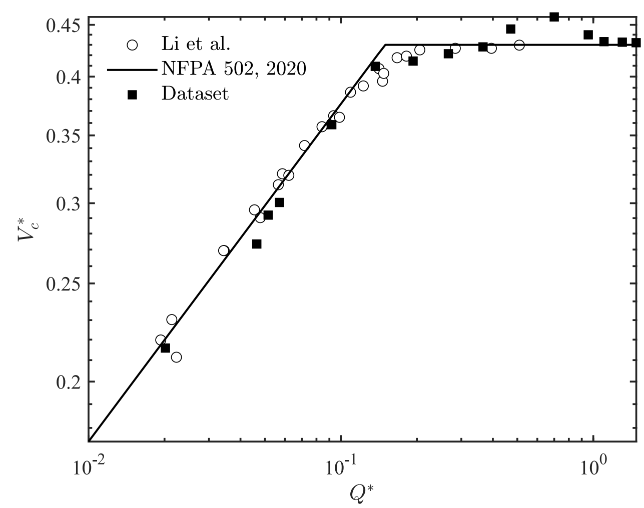

Fig. 3. Comparison of the critical velocity between the simulation data used in this work, experimental data from Li et al. (2010), and correlation proposed in NFPA (2020).

state conditions rather than the transient results, which is more statistically significant in practice. Ultimately, the data-set used in this work was simplified into 1000 different physical fields (or tensors) and 1000 sets of key physical quantities ($\dot{Q}$, θ, T_∞, v_0, H and W) that dominate the laws of tunnel fire. As this work will use the images of the physical fields as part of the input data, all of the images will be resized to the same dimension. It is the reason why H and W are not used as the input data. The simplified data-set could be thus represented by 1000 samples that denoted with $\mathbb{S} = \{\mathbf{x}, \mathbf{z}\}$, where $\mathbf{x}$ is the physical fields (temperature, smoke visibility, velocity, etc.) and $\mathbf{z} = \left(\dot{Q}, \theta, T_\infty, v_0\right)$ is the vectors with four key physical quantities. 700 samples are randomly selected as the training set, and the other 300 samples are used for the test set.

3.2. Deep learning model

To achieve the bidirectional prediction on the tunnel fire based issues, a deep learning network model is constructed as shown in Fig. 4. The lower part (colored with purple) in the figure is called the encoding part, and the upper part (colored with orange) is called the decoding part. The encoding part can reduce the physical field to several key physical quantities, so it is used for predicting the key physical quantities via physical field. The decoding part can generate the physical field from several key physical quantities, so it is used for predicting the physical field via key physical quantities. The encoding

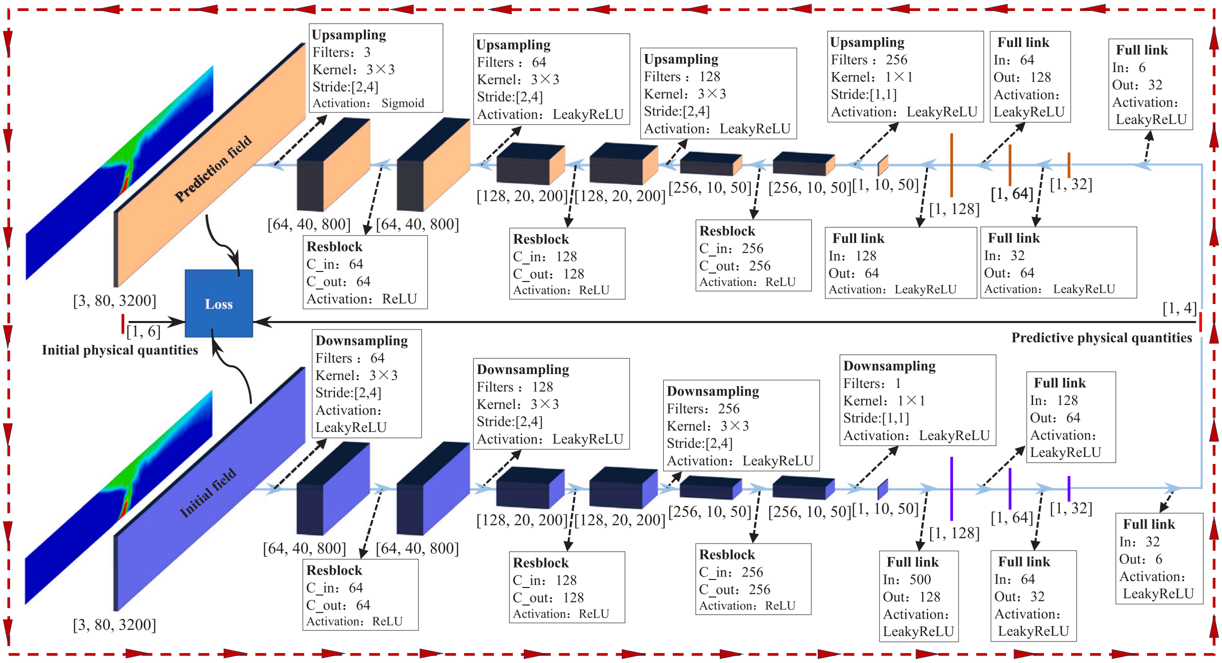

Fig. 4. Network structure of the deep learning model. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

part includes four alternating downsampling layers and residual block layers, as well as four fully connected layers. The padding size of the first three downsampling layers is 1×1, and the padding size of the last downsampling layer is 0. The size of the convolution kernel and its stride is also shown in the figure. The input tensor before the first fully connected layer needs to be reshaped into a vector of size 1×500 before the connection. The decoding part consists of four alternating upsampling layers and residual block layers, as well as four fully connected layers. The padding size of the first three upsampling layers is 1×3, and the padding size of the last upsampling layer is 1×1. The convolution kernel and its stride information can be found in the figure. The output vector of the last fully connected layer needs to be reshaped into a tensor of size $1 \times 10 \times 50$ before performing the upsampling operations. The residual blocks (Zhang et al., 2017) in the network structure play a role in preventing gradient vanishing, and each residual block has a similar structure inside, with the order: convolutional layer (with 3×3 convolution kernel, 1×1 padding, and the number of output channels is equal to the number of input channels), activation layer (ReLU function (Yarotsky, 2017)), convolution layer (with 3×3 convolution kernel, 1×1 padding, and the number of output channels is equal to the number of input channels), the stacking layer (the sum of the convolution result and the original input), and the activation layer (ReLU function). More information of the DLM can be directly read on Fig. 4.

Each sample in the training set $\mathbb{S} = \{\mathbf{x}\,(\text{initial field}), \mathbf{z}\,(\text{initial physical quantities})\}$ is seen as the input of the DLM during the training process, while the predicted field and the predicted physical quantities are seen as the output. To implement the bidirectional prediction, the loss function is naturally defined as:

$$\mathcal{L} = \underbrace{\frac{1}{3N}\sum_{c=1}^{3}\sum_{n=1}^{N}\left(x_{c,n} - x'_{c,n}\right)^2}_{\mathcal{L}_1} + k \cdot \underbrace{\frac{1}{4}\sum_{m=1}^{4}\left(z_m - z'_m\right)^2}_{\mathcal{L}_2}, \tag{1}$$

where $x_{c,n}$ is the value of nth pixel in c channel, z_m is value of the mth physical quantities in $\mathbf{z}$, k is the ratio of mean square error of $\mathbf{z}$ to $\mathcal{L}_1$. The first part of the loss function $\mathcal{L}_1$ is the deviations between the true field and the predicted field, and the second part of the loss function $\mathcal{L}_2$ is the deviations between the true physical quantities and predicted quantities. By training the DLM, all parameters of the DLM are adjusted to make the loss function as low as possible. Finally, the trained DLM could predict the physical field by the given key physical quantities ($\dot{Q}$, v_0, T_∞, and θ) of the tunnel fires via decoder part, and could predict the key physical quantities by the given physical field of tunnel fires via encoder part.

In this work, the hyper-parameters of the DLM need to be clarified are: (1) the batch size is 8; (2) the optimizer is Adam with the coefficients $\beta_1 = 0.9$ and $\beta_2 = 0.999$ (Yazan and Talu, 2017); (3) learning rate is 1.5×10^{-4}, the low order number used to prevent the denominator from being zero is 1.5×10^{-8}, the training epoch is 500.

The construction, training, and test of the proposed DLM are implemented in the System of Rocky Linux v8.7 with Pytorch v2.1.2 and Python v3.12. The GPU used for training is NVIDIA TESLA V100.

3.3. Data processing

To have an easier way to train the DLM and converge the loss function, the input and output data of the DLM are processed and preprocessed. Fig. 5 illustrates the methods of data preprocessing. The step 'time-averaging' in the figure means averaging the transient physical fields to the time-averaged physical fields, which follows the explanation in section 3.1. However, the inclination of the tunnel leads to a large number of 'white' pixels in the rectangle image. To avoid invalid information, we rotate the image to the horizontal visually and trim off the white pixels (step 'rotating to the horizontal' in Fig. 5). Next, to enable the model can train all images of physical fields with different tunnel dimensions, the input size of the images is strictly resized to 1600 (horizontal) × 80 (vertical) × 3 (channels). Before being inputted into the encoder, the resized images are normalized to reduce the difficulties of model training. Also, the output of the encoder should be normalized inversely, and its normalized result is the final predicted key physical quantities. These processes describe the data preprocessing in the encoder part of the DLM. The data processing in the decoding process is similar to the above process, except that the above processes are performed in reverse. It is noted that the normalization is used multiple times because it needs to be performed once every time the data passes through the encoder and decoder, as seen in Fig. 5. The specific mathematical expressions of the normalization and inverse normalization can be found in the Fig. 5.

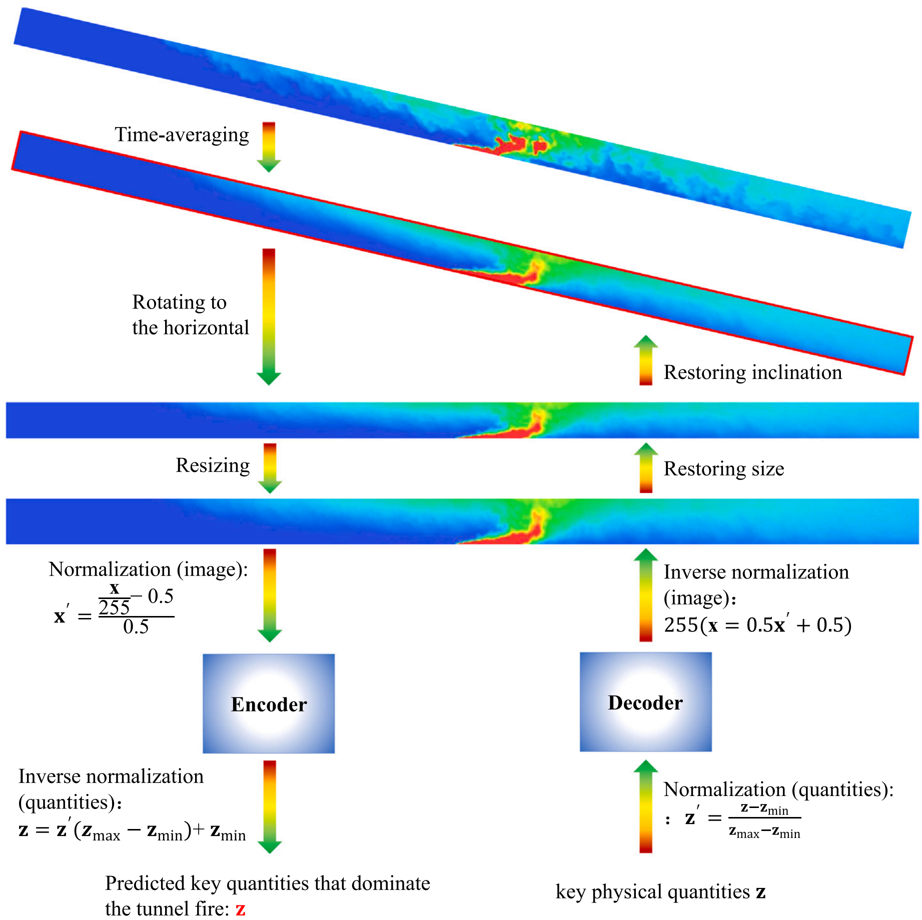

Fig. 5. Schematic diagram of data processing and preprocessing.

4. Results and discussion

4.1. Training of the prediction model

In the loss function Eq. (1), $\mathcal{L}_1$ and $\mathcal{L}_2$ are respectively calculated by comparing the deviations of the field images and the physical quantities, their ratio k may significantly affect the prediction performance. By taking the prediction between the temperature field and the four key physical quantities ($\dot{Q}$, θ, T_∞, v_0) as the example, Table 1 lists the R^2 of the test dataset which calculated by the trained DLMs with different ratios k. The result shows that $k = 200$ get the best performance with the highest average R^2 0.865, where R^2 of T_∞, $\dot{Q}$, and v_0 are all over 0.92. The inclination angle θ does not seem to get high scores of R^2 for all ratios in listed in Table 1, they are less than 0.6 except in the case of $k = 200$. This result indicates the θ is not a quantity that could be easily and accurately predicted when the temperature field is known, but the T_∞, $\dot{Q}$, and v_0 can be well predicted. As a result, the ratio $k = 200$ is chosen as the basic in the context below.

Next, the evolutions of the loss function are plotted in Fig. 6 to know what happened during the training process. It is seen that the $\mathcal{L}$, $\mathcal{L}_1$, $\mathcal{L}_2$, and $\mathcal{L}_{test}$ are all decay with the train epoch, and almost converged after the sixtieth epoch. In addition, it is observed that $\mathcal{L}$ is dominated by the $\mathcal{L}_1$ before the sixtieth epoch, whereas $\mathcal{L}_1$ and $\mathcal{L}_2$ have similar contributions later. Moreover, $\mathcal{L}_{test}$ is higher than $\mathcal{L}$ in the whole training process, which reveals that the DLM is somehow over-fitted. In fact, since the image prediction contains three channels for a large number of pixies, complex DLMs with massive parameters are prone to over-fitting. This may hint to us to reduce some model complexity. However, the over-fitting could be accepted if the $\mathcal{L}_{test}$ gets the continuous decay during the training process and the prediction performance is sound. Therefore, the complexity of the DLM is currently not discussed before the specific prediction performance is tested (acceptable performance can be actually seen below).

Table 1
R^2 of the test data calculated by the deep learning model with different ratio k.

k	R^2				
	T_∞	$\dot{Q}$	v_0	θ	Average
5	0.956	0.898	0.9575	0.594	0.851
10	0.912	0.865	0.950	0.518	0.811
20	0.943	0.906	0.9468	0.562	0.839
50	0.960	0.886	0.960	0.554	0.840
100	0.962	0.896	0.953	0.591	0.851
200	0.957	0.922	0.9524	0.631	0.865
300	0.953	0.908	0.955	0.528	0.836

Fig. 7 displays the enhancement of the prediction ability of the DLM on predicting the key physical quantities T_∞, $\dot{Q}$, v_0, and θ during the training process. The shown result is a case randomly chosen from the training set, but it has typical representativeness for the whole training set. In the figure, the red lines represent the true value of the key physical quantities, and the blue curves represent the predicted values during the training process. It can be seen that all quantities do not approach their true values in the early stages of the training process because the DLM is still 'naive' at that time. When the training epoch reaches around 100, all quantities almost converge to their true values. Although some slight oscillations are observed after even the hundredth training epoch, their mean relative errors (calculated from

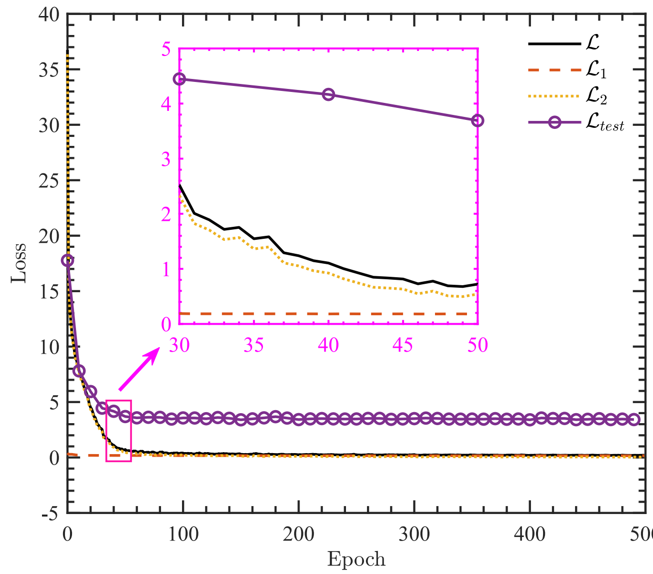

Fig. 6. Evolution of the loss functions on the train dataset and test dataset.

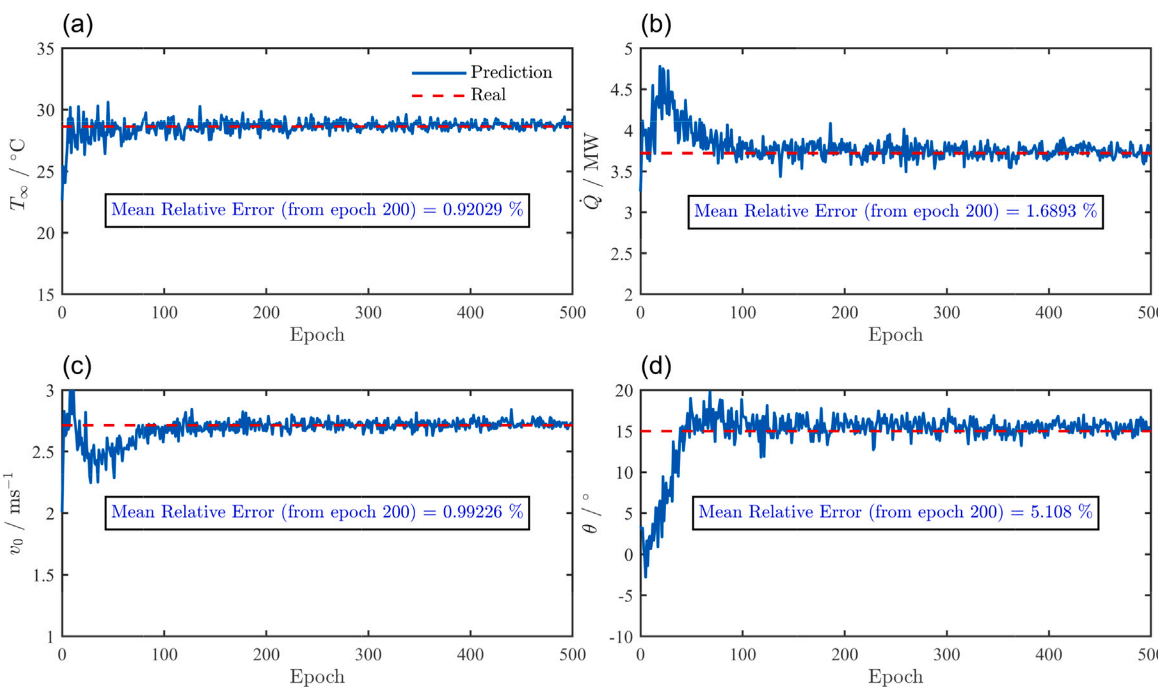

Fig. 7. Evolution of the (a) T_∞, (b) $\dot{Q}$, (c) v_0, and (d) θ that are predicted by taking the temperature as the input during the training process. This result is a case randomly selected from the training dataset.

the two-hundredth epoch) are low: 0.92029%, 1.6893%, 0.99226% and 5.108% for T_∞, $\dot{Q}$, v_0 and θ, respectively. Again, the θ seems to not be 'obedient' compared with other quantities because its values are still decaying from the two-hundredth epoch, which means its value is easier to be over-fitted and is more difficult to predict precisely. This may be because the temperature field is not mainly dominated by the inclination angle (within the scope discussed in this work at least), so it is not easy to estimate the inclination angle from the morphology of the whole temperature field. Nevertheless, the prediction results of the four key quantities overall show good learning ability of the DLM from the training set.

Fig. 8 shows the calculated temperature fields by the DLM during the training process. Noted that the two cases are randomly chosen from the training set. It is seen that the DLM does not have any prediction ability at the beginning: the calculated fields does not capture any changes in the temperature in the whole tunnel. After the tenth train epoch, the model can roughly predict the contours of the high-temperature smoke and flame, but the temperature fields around the flame are still in 'immature' states. When the training epoch reaches 200, it can be seen that the model has good predictive ability: because predicted temperature fields are very similar to the true temperature fields. When the training epoch reaches 500, although the predicted results do not perform as smoothly as the true temperature field at the outer contours of the 'smoke', the calculated temperature fields and the true temperature fields are almost consistent. As a result, it can be confirmed that the DLM continuously learned the field information with the increase of train epoch.

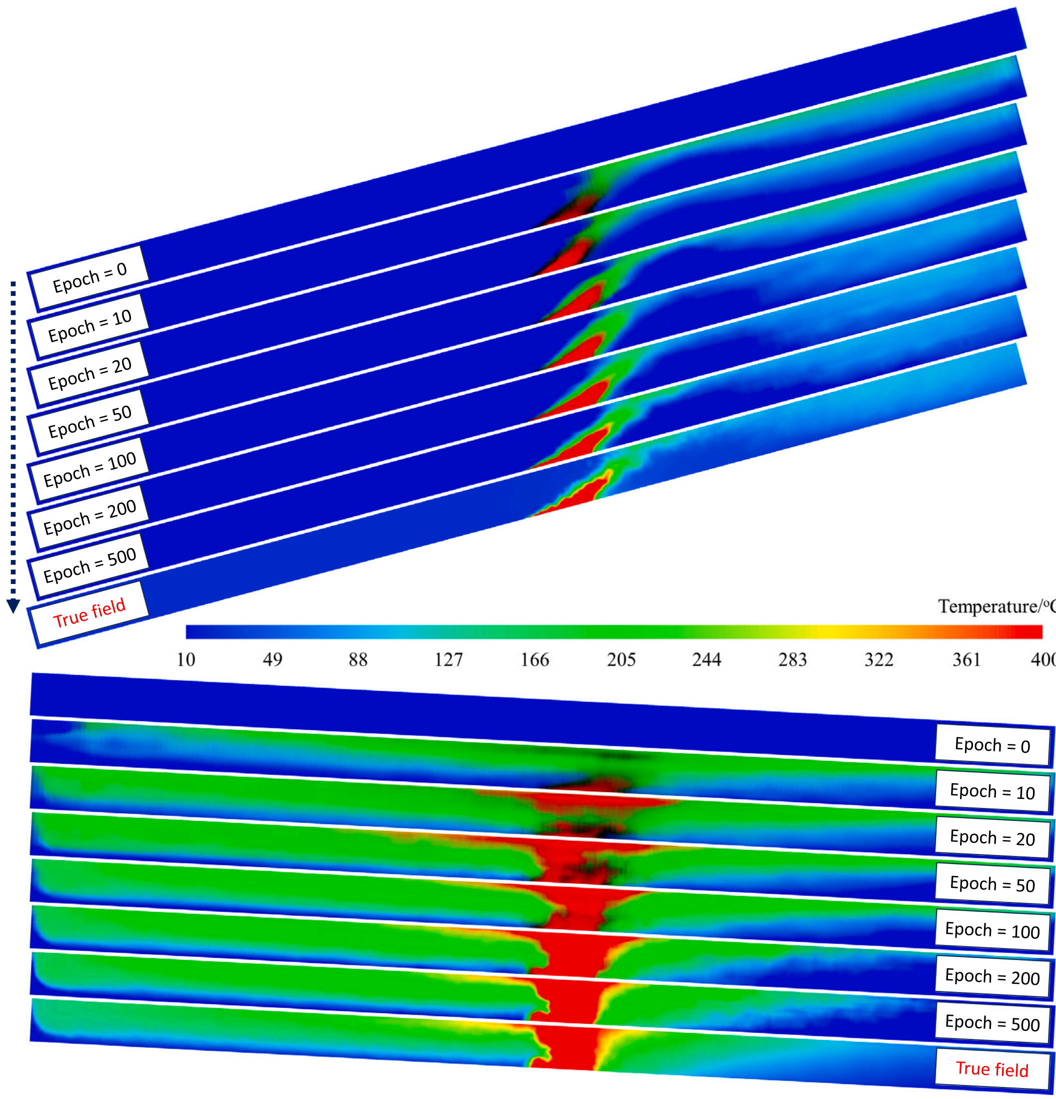

Fig. 8. Evolutions of the temperature field predicted by taking the T_∞, $\dot{Q}$, v_0, and θ as the input during the training process. These two cases are randomly selected from the training dataset.

4.2. Prediction performance

The learning ability of the model on the training set has been demonstrated in the previous section, but it is still necessary to further evaluate the predictive performance of the model through the test set after the model training is completed. To test the prediction performance from key physical quantities to the physical fields, Fig. 9 selects some cases from the 300 test cases for comparison. To well show the results, these selected cases are sorted from the low inclination angle to the high inclination angle. The left side is the true temperature fields and the right side is the predicted temperature fields. It is noted that for visually easy comparison, the predicted temperature fields on the right side have been symmetrically transformed and the color bar range of these temperature fields is limited between 10 °C and 400 °C. Therefore, the good symmetry of the results means an excellent prediction performance. It should be emphasized these selected cases have different key physical quantities (the different tunnel heights and inclination angles in the figure clearly indicate this), so the result also indicates that the DLM has strong robustness for predicting the temperature field under different geometric conditions.

Fig. 10 is another example that predicting the visibility field caused by the fire smoke. The sorting rule of these cases is consistent with Fig. 9. The network structure and the loss function employed in this prediction are the same as in Section 4.1. It can be seen that some details are different between the true visibility field and the predicted visibility field, e.g., several obvious discontinuity breaks in the first, fourth, seventh, and ninth predicted visibility fields. This indicates that the prediction cannot reproduce all the detailed contours of the smoke. However, the symmetry is well performed from the entire Fig. 10, which overall illustrates the good prediction on the visibility field.

In short, it can be seen that deep learning models have good potential to predict various physical fields in tunnel fires through key physical quantities. Such predictions are made instantaneously, which is more efficient than directly simulating the temperature fields, visibility fields, etc. Therefore, the proposed method may provide some convenience for prediction in tunnel fires.

Looking back at the prediction performance on the key physical quantities by using the physical fields. Fig. 11 illustrates the predicted T_∞, $\dot{Q}$, v_0 and θ from test set. The result is predicted by taking the temperature fields as the input of the DLM. The standard deviations (σ) of the predicted T_∞, $\dot{Q}$, v_0 and θ are 1.155 °C, 0.356 MW, 0.173 m s^{-1} and 4.224°, respectively. The 3σ limits could almost cover all the predicted results. In addition, the mean relative errors (ϵ) of the predicted T_∞, $\dot{Q}$, v_0 and θ are 5.251%, 12.879%, 12.031% and 92.235%, respectively. Combining the results shown in Table 1, all these statistical results imply the ambient temperature, HRR, and ventilation velocity have

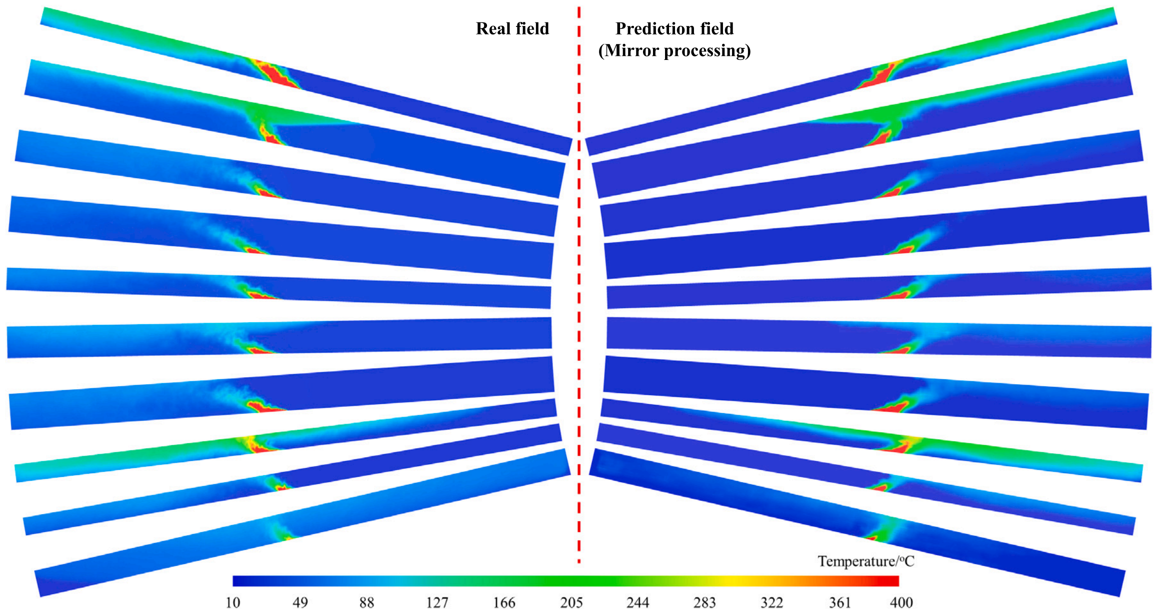

Fig. 9. True temperature fields vs. the predicted temperature fields. These are randomly selected cases from the test set and sorted by the values of the inclination angle. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

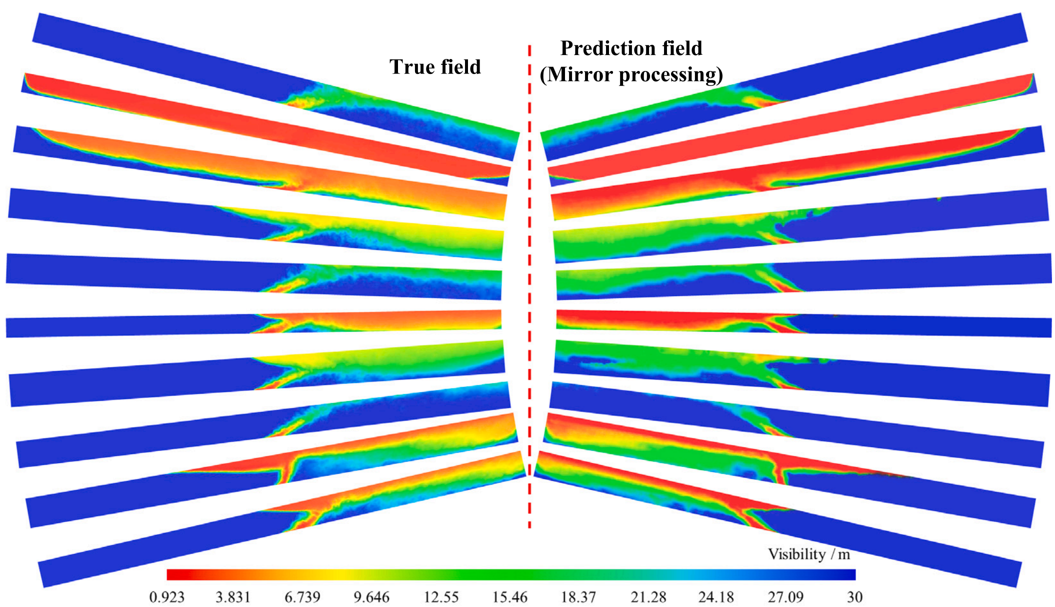

Fig. 10. True visibility fields vs. the predicted visibility fields. These are randomly selected cases from the test set and sorted by the values of the inclination angle.

higher chances of being accurately predicted, whereas the prediction accuracy of the inclination angle is lower. It is may because the images of the physical field are rotated before they are used for training, which leads some visual information for the DLM being lost. Fortunately, the prediction of the other three physical quantities is relatively accurate, which means the proposed model can still provide a way for accident inversion investigations after tunnel fires occur.

Moreover, Fig. 12 presents the predicted physical quantities by using the visibility field as the input of the DLM. Unlike the result in Fig. 11, the predicted physical quantities have higher standard deviations and mean relative errors. Especially for the predicted T_∞ and θ, the σ and ϵ are respectively over than 30% and 180% for the predicted T_∞ and θ. The R^2 of T_∞ and θ were calculated as well, their values are respectively −4.626 and 0.419. This indicates that no positive correlation is observed between the predicted ambient temperature and true ambient temperature, and some positive correlation is observed between the predicted inclination angle and the true inclination angle even though the error is high. The reason is obvious: the color remains the same in the image of the visibility field as long as there is no smoke, so it does not contain any information about the ambient temperature. On the contrary, the temperature field image itself contains ambient temperature information, because the display range of the color bar is limited between 10 and 400 degrees in the images. Ventilation velocity and HRR have better predictive performance, even though they are still slightly inferior to the prediction performance by using temperature fields.

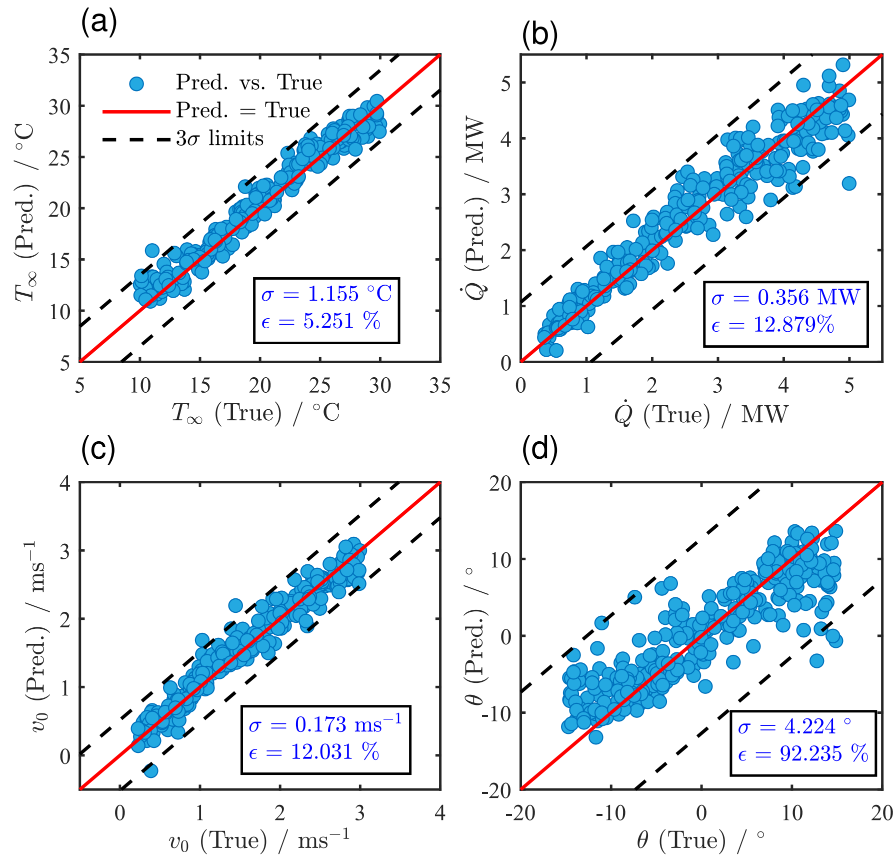

Fig. 11. Predicted physical quantities by the temperature field vs. true physical quantities. σ and ϵ denote respectively the standard deviation and mean relative error.

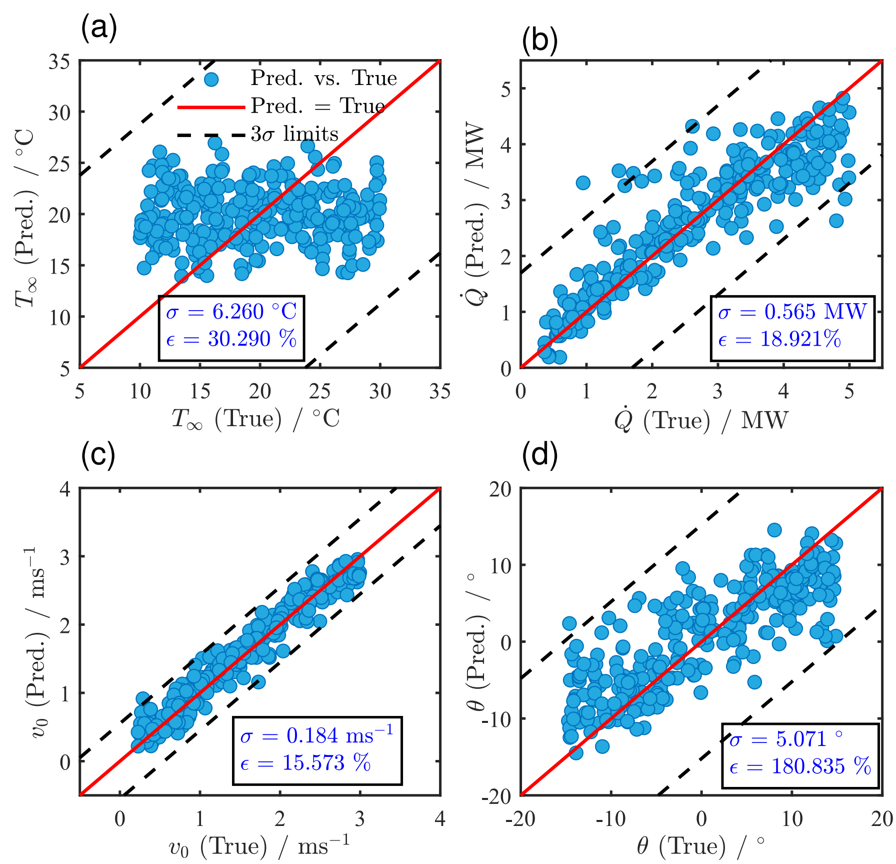

Fig. 12. Predicted physical quantities by the visibility field vs. true physical quantities. σ and ϵ denote respectively the standard deviation and mean relative error.

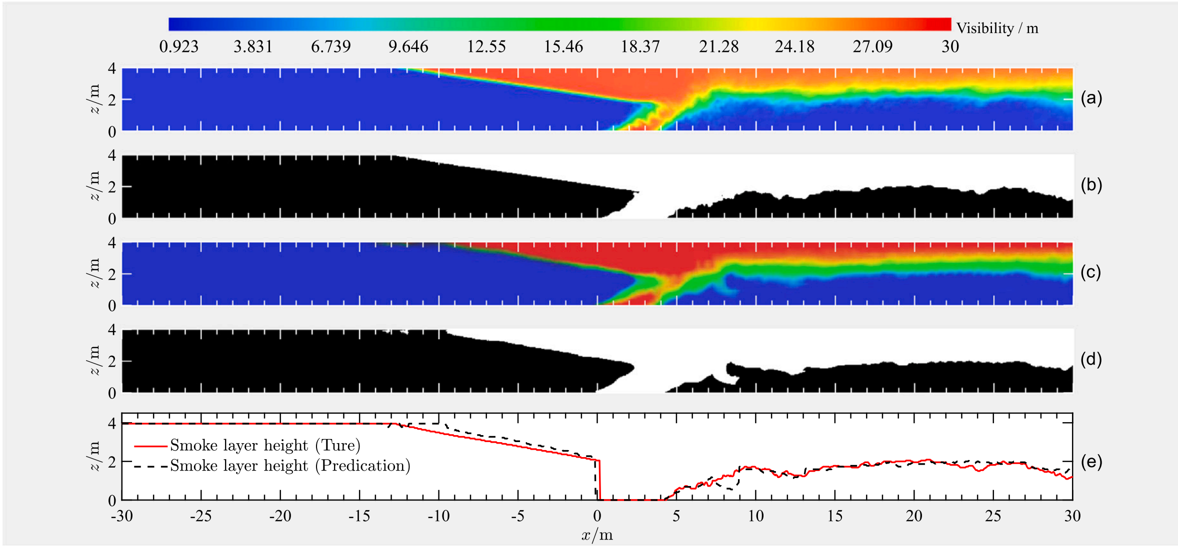

Fig. 13. The (a) true temperature field and the (c) predicted temperature field, as well as their binarized images (b, d). (e) is the smoke layer height calculated from (b) and (d). This is a case randomly selected in the test set.

4.3. Prediction on other important parameters

In practice, some important parameters play a crucial role in the firefighting and rescue decision-making of tunnel fires, such as the distribution of smoke layer height and the smoke back-layering length. Fig. 13 shows a case with the true visibility field and the predicted visibility field, as well as their binarized images. The shown case is chosen from the test set. Here, the binarized images are obtained by converting color images to grayscale images (Song et al., 2013). The conversion calculation formula developed here for this type of issue is

$$a_{i,j} := \begin{cases} 1, a_{i,j} > \frac{4}{5MN}\sum_{i=1}^{M}\sum_{j=1}^{N} a_{i,j} \\ 0, a_{i,j} \le \frac{4}{5MN}\sum_{i=1}^{M}\sum_{j=1}^{N} a_{i,j} \end{cases}, \tag{2}$$

where $a_{i,j}$ is the pixel value in the ith row and jth column of the image, M is the total rows of the image pixels, N is the total columns of the image pixels. The binarized images are used for calculating the distribution of smoke layer height and the smoke back-layering length via horizontal and vertical scanning methods:

$$h(x) = \left(1 - \frac{\max\left\{i : (i = 1, 2, 3, \ldots, M)\left(a_{i,\lceil \frac{N}{2} + \frac{xN}{L} \rceil} = 1\right)\right\}}{M}\right) H \tag{3}$$

and

$$l = \frac{N - 2\min\left\{j : (j = 1, 2, 3, \ldots, N)\left(a_{1,j} = 1\right)\right\}}{2N} L \tag{4}$$

where $h(x)$ is the smoke layer height along the x-direction, l is the back-layering length, L is the length of the tunnel, $\lceil\cdot\rceil$ is the ceiling function. Fig. 13(e) is the calculated smoke layer height from Fig. 13(b) and Fig. 13(d), respectively. Overall, the deviation between the smoke layer height distributions calculated by the predicted fields and true fields is low.

To quantitatively evaluate the prediction performance, Fig. 14 shows the frequency histograms of errors between the predicted result and the true result: $\varepsilon_{SH} = h_t - h_p$, where h_t and h_p are respectively the true smoke layer height and the predicted smoke layer height point by point along the x-axis; $\varepsilon_{BL} = l_t - l_p$, where l_t and l_p are respectively the true back-layering length and the predicted back-layering length. The mean absolute errors, $\overline{|\varepsilon_{SH}|}$ and $\overline{|\varepsilon_{BL}|}$, are respectively 0.124 m and 1.432 m. The distribution of ε_{SH} has more characteristics close to the normal distribution. Approximately 70% and 65% ε_{SH} and ε_{BL} are both around 0. Moreover, the R^2 calculated by h_t and h_p is 0.95, and is 0.92 if calculated by l_t and l_p. The results demonstrate a good potential for predicting key parameters of tunnel fires as well.

5. Challenges, limitations, and considerations

This work demonstrates the potential for bidirectional rapid prediction for tunnel fires, but further considerations and limitations need to be explained in practice. Firstly, although the proposed model has shown good performance within the considered parameters and their variation ranges, actual tunnels may have more complex ventilation systems, geometric shapes, and personnel interactions. Therefore, one of the future challenges is to continuously expand the scope and scale of the database to enhance the capacity of the proposed model in various scenarios. A software can also be developed and upgraded according to the expansion of the database to aid in fire design and fire accident investigation. In addition, rapidly predicting the entire physical field of tunnel fire from key control parameters is helpful for revealing the distributions of the physical quantities in tunnel fires. However, it is relatively complex to predict the key control parameters from the physical field, as it is challenging to obtain the complete physical field data of the tunnel fire. Here are three ideas to help solve this issue: (1) Install sufficient sensors in the tunnel and utilize the multidimensional linear interpolation method to obtain an approximate physical field, and then use the proposed DL model to predict key control parameters in tunnel fires; (2) Install a small number of sensors at key locations in the tunnel, and use the data obtained by the sensors to predict other parameters, as the work of Wu et al. (2020).; However, due to the small number of sensors, this method may not be as effective as the idea (1); (3) Install a certain number of infrared stereo cameras in the tunnel and obtain the physical field of the tunnel fire as much as possible through data fusion, but strict attention should be paid to the temperature resistance of the cameras. In the future, all the above considerations and challenges need to be focused on in practice.

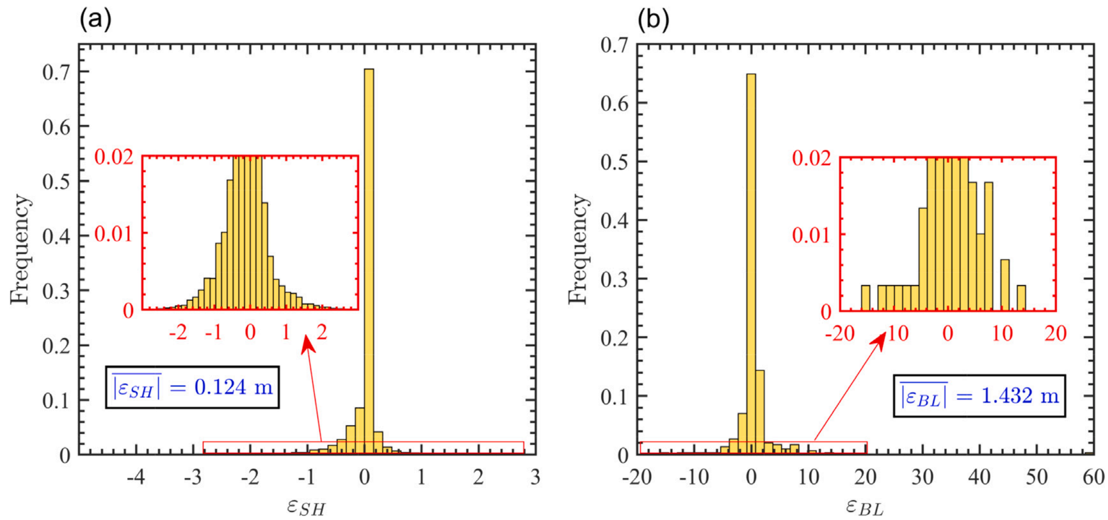

Fig. 14. (a) The frequency histograms of errors between the predicted smoke-layering height and true smoke-layering height, as well as (b) the frequency histogram of errors between the predicted back-layering length and true back-layering length.

6. Conclusions

In this paper, a prediction frame based on the DLM was developed for the purpose of bidirectional prediction between the full-field information and a few key physical quantities in the tunnel fire. By establishing a large numerical database containing 1000 sets of tunnel fire simulation results with high-resolution grids, the strategies of the data preprocess and data process were proposed to generate the training set and test set. The loss function of the DLM was determined by comparing the prediction performance under different weights in terms of all sub-terms in the loss function.

The proposed DLM was trained and its loss function converged at the eightieth training epoch for both the training set and the test set. With the increase of the training epoch in the training process, the four key physical quantities (ambient temperature, HRR, ventilation velocity, and inclination angle) calculated by the encoder part of the DLM were gradually close to their true value and converged at the hundredth train epoch. On the other hand, the decoder part of the DLM can output the calculated physical fields that were increasingly close to the true physical fields, and it is almost identical to the true physical field by the hundredth train epoch.

The trained DLM was used for prediction and was evaluated by the test set. The results showed that the model can predict the full temperature and visibility fields instantly through several key physical parameters, though the geometric and thermal conditions are not known. Conversely, the key physical parameters can be predicted through the full physical fields as well. The ambient temperature, HRR, and ventilation velocity were well predicted by the full temperature fields, and the tunnel inclination angle was relatively poor. When using the visibility fields for prediction, the ambient temperature cannot be predicted.

The predictive capabilities of the proposed model on the two practical important parameters, the smoke layer distribution and smoke back-layering length were evaluated by the generated physical fields. The mean absolute errors of the predicted two practical important parameters are respectively 0.124 m and 1.432 m, and the R^2 of them are respectively 0.95 and 0.92.

CRediT authorship contribution statement

Yao Hong: Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization. **Congling Shi:** Writing – review & editing, Supervision, Resources. **Fei Ren:** Writing – review & editing, Funding acquisition. **Xiaohu Wu:** Writing – review & editing, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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